

A PRELIMINARY SURVEY OF HUMAN MORPHOGENETIC TRAITS WITH SIMPLE INHERITANCE PATTERNS IN GRADUATE STUDENTS OF SKR AND SKR GOVERNMENT COLLEGE FOR WOMEN, KADAPA, ANDHRA PRADESH, INDIA

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This work was carried out in collaboration between both authors. Both authors read and approved the final manuscript.

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ABSTRACT

Many factors such as gene flow, genetic drift, migration, natural selection and temporal variation are responsible for the genetic variation in humans. The present pilot study was conducted to screen a random population of 194 female Indian graduate students for 13 different morphogenetic traits i.e., tongue rolling, tongue folding, polydactyly, hitchhiker's thumb, bent little finger, widow's peak, cleft chin, dimpled cheeks, mid-phalangeal hair, earlobe attachment, crossing of arms, right handedness and arched foot through questionnaires and datasheets. The prevalence of these morphogenetic traits was observed as dominant traits such as tongue rolling ability (75.3%) and tongue folding ability (96.4%), single-jointed hitchhiker's thumb (76.3%), straight little finger (93.3%), free ear lobe (62.4%), arched foot (88.7%), right handedness (99%), right arm crossing over left arm (99.5%) etc. are expressed more recurrently in the population. However, recessive characters like absence of polydactyly (97.9%), absence of dimples (87.6%), absence of widow's peak (77.8%), round chin (87.1%), absence of mid-phalangeal hair (77.1%), etc. are expressed more commonly in the population. We analyzed whether a trait is dominant or recessive. The survey result shows a disparity of this argument, it shows that some typical dominant character was not expressed but the expression of recessive character was prevalent as in the case of widow's peak (absent), smooth chin,

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crossing of arms (right over left), Hitchhiker's thumb (absent), absence of dimple, etc, all of which are recessive traits. Outcomes of this study can be suitable for many types of anthropological, medicolegal and population genetics studies.

Keywords: Dominant; frequency; genetic variation; population; prevalence; recessive; traits.

1. INTRODUCTION

Morphogenetic traits are the somatic features of an individual, which are inherited as autosomal dominant and autosomal recessive traits [1,2,3], and cause the genetic variance among the living organisms [4]. Human populations serve as a best specimen to study the morphogenetic variations among the different populations living in different ecological and geographical conditions [5]. There are many aspects such as gene flow, genetic drift, independent assortment which influence the genetic variations among the individuals [6]. Genes are responsible for the expression of the traits in the individuals [7], and these traits are inherited to the next generations by dominant and recessive alleles of one gene and are designated as 'Mendelian traits' [8]. However, some traits are considered to be Non-mendelian as their alleles are located on more than one gene and are considered as 'Polygenic traits' such as height, weight, hitchhiker's thumb, bent little finger and handedness with varying degrees of variation [9].

The incidence of a trait in a population can be associated to whether its phenotypic effect is favourable or unfavourable. Usually there is a misapprehension that dominant alleles are the most frequently occurring alleles i.e., favourable traits, and they tend to suppress and eliminate the recessive alleles i.e., unfavourable traits in due course of time [4]. But, the survival rates of the dominant and recessive traits depend on the genetic expression of this trait.

The study of inheritance pattern of morphogenetic traits among the population is very necessary as it helps to analyse the genetic history of a population and determines the chances of occurrence of genetic diseases in the society. The present study focused on some human traits which are more prevalent in female population of Kadapa District, Andhra Pradesh. The female students as a study group was selected because of the presence of diversified communities in the college. It was only a preliminary attempt to assess the expression of some frequently occurring traits in a female population of Sri Kotireddy and Srimathi Kotireddy Ramasubbamma (SKR and SKR) Government College for Women, Kadapa, Andhra Pradesh, India. This pilot study checked which trait is more dominantly or recessively expressed among the female population, to make a comparison of religion

percentage difference in the expression of the traits and find out the association among the different traits.

2. MATERIALS AND METHODS

2.1 Study Population

The pilot survey was conducted in randomly selected female students between the age group of 17-24 years of SKR and SKR Government College for Women, Kadapa district (14°28'01.7"N 78°49'25.5"E).

2.2 Data Collection

The survey method was chosen for data collection with a questionnaire and the data was analyzed through a datasheet including the individual's name (identity will not be disclosed), age, religion, trait based on the phenotypic expression in that individual (Appendix-I). A total of 194 female individuals were studied for 13 different morphogenetic traits such as tongue rolling, tongue folding, polydactyly, hitchhiker's thumb, bent little finger, widow's peak, cleft chin, dimpled cheeks, mid-phalangeal hair, earlobe attachment, crossing of arms, right handedness and arched foot from the random population. These morphogenetic traits were selected due to their pattern of inheritance as autosomal dominant and autosomal recessive traits and the ease to observe them in the sampled population. Following are the description of selected morphogenetic traits.

1. **Tongue rolling:** The capability to turn up the lateral edges of the tongue into U shape is controlled by a single dominant gene [10].
2. **Tongue folding:** The ability to fold the tip of the tongue is controlled by a single dominant gene [10].
3. **Polydactyly:** Polydactyly is a condition where an individual has extra fingers or toes, caused by the dominant allele of a gene [11].
4. **Hitchhiker's thumb:** Thumb can be straight or curved (Hitchhiker's thumb). A curved thumb is dominant over a straight thumb. Straight thumbs can be seen as nearly a straight line and may contain a slight arch when viewed from the side [12].
5. **Bent little finger:** Little finger can be straight or bent. A bent little finger occurs due to the expression of a dominant allele where the end

joint of the little finger of each hand is bent inwards [12].

6. **Widow's peak:** A widow's peak or the mid-digital the hairline is due to expression of the dominant allele of the hairline gene and straight hairline occurs due to the expression of a recessive allele [13].
7. **Cleft Chin:** A cleft is a dimple on the chin which is expressed due to a dominant gene while the recessive genotype presents without a cleft or round chin [14].
8. **Dimple cheeks:** Dimples are round depressions in the cheeks when laughing, not lines or clefts which are expressed due to a single dominant gene [15].
9. **Mid-phalangeal hair:** The presence of hair on the back of the middle segment of the finger, excluding thumb is expressed by dominant gene, while its absence is due to the expression of a recessive gene [16].
10. **Earlobe attachment:** Ear lobes may be free or attached in an organism. The attached ear lobe is expressed by a recessive gene while the free ear lobe is expressed due to a dominant allele [17].
11. **Crossing of arms:** Arms folded across the chest, with the left arm placed on top is the dominant characteristic, while the right arm positioned on the top is the recessive characteristic [18].
12. **Right handedness:** Right-handedness is inherited by the dominant gene and majority of the people show this trait and the gene for left hand is recessive [19].
13. **Arched foot:** The presence of an arch, i.e., a distinct curve along the inside of the foot with a band slightly less than half the width of the foot connecting the heel and toe is due to the expression of a dominant allele, while the allele for flat foot is recessive and shows a change in foot shape in which the foot does not have a normal arch when standing [4].

2.3 Statistical Analysis

After the primary data collection, the data was consolidated and statistical analysis was performed using SPSS software and MS-Excel. Evaluated traits were grouped based on location i.e. on the face (widow's peak, cleft chin, earlobe attachment and dimpled cheeks), tongue (tongue rolling and tongue folding), fingers (bent little finger, hitchhiker's thumb, mid-phalangeal hair, polydactyly) and hands and feet (crossing of arms, right handedness, arched foot) and were analyzed for their associations using chi-square test of independence. A comparison of the prevalence of the selected trait expressions within religion and

population was determined by using chi-square analysis. Values were considered significant at $p < 0.05$.

3. RESULTS

3.1 Distribution of Dominant and Recessive Expressed Phenotypes of Morphogenetic Traits

Out of 194 individuals, 146 (75.2%) were capable of rolling their tongue and 187 (96.3%) were capable of folding their tongue. Only, 4 (2.06%) individuals showed polydactyly. A single jointed thumb was observed in 148 (76.2%) individuals of the sampled population. 181 (93.2%) samples showed a straight little finger. Only, 43 (22.2%) subjects showed widow's peak (Curved hairline) in their head. The results showed that only 25 (12.8%) individuals had cleft chin. The analysis showed that only 24 (12.3%) subjects of the sampled population showed facial dimples. Only 56 (28.2%) subjects showed the mid-phalangeal hair on their fingers. A total of 121 (62.3%) samples showed free earlobes. The pattern of crossing of arms were surveyed in which 193 (99.4%) individuals cross their arms right over left. Only 2 individuals have left handedness out of 194 subjects. Only, 172 (88.6%) individuals are having arched foot (Fig-1; Table-1).

3.2 Distribution of Morphogenetic Traits Amongst Religions

Among the 194 sampled individuals, only 19 subjects were Christian (9.8%), 129 (63.4%) subjects were Hindus, and the remaining 52 (26.8%) subjects were Muslim (Table-1). Only, 68.4% Christians, 76.4% Hindus and 75% Muslims were able to roll their tongue (Table-2). A majority of the population among the three religions showed the ability to fold the tongue (100% Christians, 96.7% Hindus and 94.2% Muslims), and the absence of polydactyly (100% Christians, 98.3% Hindus and 96.1% Muslims). The survey showed that 73.6% Christians, 75.6% Hindus and 78.8% Muslims showed a single jointed hitchhiker thumb (straight thumb). Only, 5.2% Christians, 7.3% Hindus and 5.7% Muslims had a bent little finger. In the present survey, 15.7% Christians, 21.9% Hindus and 25% Muslims had a curved hairline (Widow's peak). Similarly, 21% Christians, 13% Hindus and 9.6% Muslims had a cleft chin. Likewise, 21% Christians, 21.9% Hindus and 25% Muslims had facial dimples. Correspondingly, 52.6% Christians, 73.9% Hindus and 71.1% Muslims showed the absence of a mid-phalangeal hair. Most of the sampled population showed right over the left type

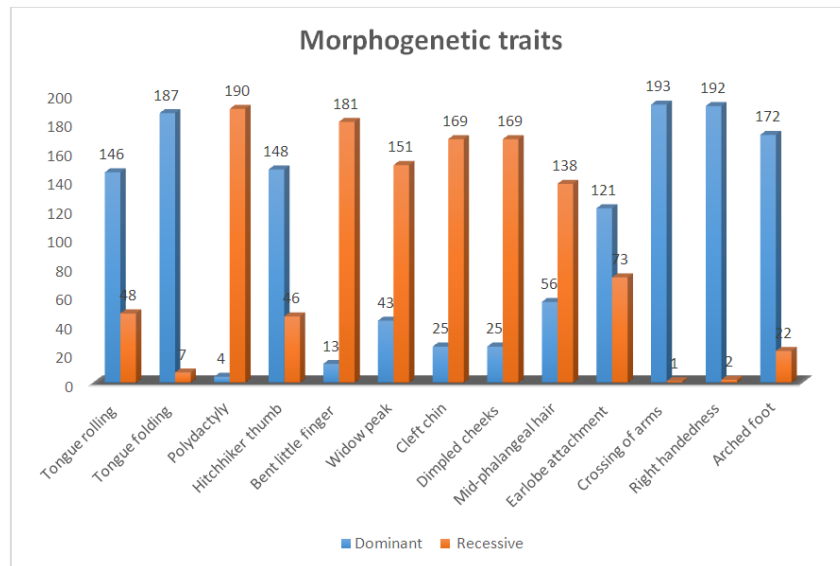


Fig. 1. Distribution of dominant and recessive expressed phenotypes of morphogenetic traits

Table 1. Descriptive statistics of traits in the sampled female population of study

Traits	Frequency (N)	Percentage (%)
Religion		
Christian	19	9.8
Hindu	123	63.4
Muslim	52	26.8
Tongue rolling		
Rollers	146	75.3
Non-rollers	48	24.7
Tongue folding		
Folders	187	96.4
Non-folders	7	3.6
Polydactyly		
Present	4	2.1
Absent	190	97.9
Hitchhiker's thumb		
Single jointed	148	76.3
Double jointed	46	23.7
Bent little finger		
Bent	13	6.7
Straight	181	93.3
Widow's peak		
Curved hairline	43	22.2
Straight hairline	151	77.8
Cleft Chin		
Cleft	25	12.9
Round	169	87.1
Dimpled cheeks		
Present	25	12.9
Absent	169	87.1
Mid-phalangeal hair		
Present	56	28.9
Absent	138	71.1
Ear lobe attachment		

Traits	Frequency (N)	Percentage (%)
Attached	73	37.6
Free	121	62.4
Crossing of arms		
Right over left	193	99.5
Left over right	1	0.5
Right handedness		
Right	2	1.0
Left	192	99.0
Arched foot		
Arched	172	88.7
Flat	22	11.3

Table 2. Test of independent association between religion and estimated traits in the population of study

Traits	Phenotype	Religion			χ^2	P at 5% level of significance
		Christian	Hindu	Muslim		
Tongue rolling	Rollers	13	94	39	0.568	0.75 (NS)
	Non-rollers	6	29	13		
Tongue folding	Folders	19	119	49	1.45	0.483
	Non-folders	0	4	3		
Polydactyly	Present	0	2	2	1.33	0.512
	Absent	19	121	50		
Hitchhiker's thumb	Single jointed	14	93	41	0.29	0.865 (NS)
	Double jointed	5	30	11		
Bent little finger	Straight	18	114	49	0.21	0.900 (NS)
	Bent	1	9	3		
Widow's peak	Curved hairline	3	27	13	0.69	0.71 (NS)
	Straight hairline	16	96	39		
Cleft Chin	Cleft	4	16	5	1.63	0.44 (NS)
	Round	15	107	47		
Dimpled cheeks	Present	4	14	6	1.46	0.48 (NS)
	Absent	15	109	46		
Mid-phalangeal hair	Present	10	91	37	3.65	0.161 (NS)
	Absent	9	32	15		
Ear lobe attachment	Free	12	71	38	3.67	0.159 (NS)
	Attached	7	52	14		
Crossing of arms	Right over left	19	122	52	0.58	0.75
	Left over right	0	1	0		
Right handedness	Right	19	122	51	0.66	0.72
	Left	0	1	1		
Arched foot	Arched	14	112	46	4.94	0.084 (NS)
	Flat	5	11	6		

Table 3. Test of independence association between tongue traits

Trait	Phenotype	Tongue rolling		χ^2	P
		Rollers	Non-rollers		
Tongue folding	Folders	143	3	4.09	0.043 (S)
	Non-folders	44	4		

Table 4. Test of independence association between facial traits

Trait	Phenotype	Cleft chin		Ear lobe attachment		Dimple Cheeks	
		Cleft	Round	Attached	Free	Present	Absent
Widow's peak	Curved	7	36	30	13	1	42
	Straight	18	133	91	60	22	129
		$\chi^2 = 0.566$ p= 0.452 (NS)		$\chi^2 = 1.288$ p= 0.256 (NS)		$\chi^2 = 4.80$ p= 0.28 (S)	
Cleft Chin	Cleft	-	-	18	7	5	20
	Round	-	-	103	66	19	150
Earlobe attachment	Free	-	-	-	-	14	107
	Attached	-	-	-	-	10	63
						$\chi^2 = 0.1903$ p= 0.663 (NS)	

Table 5. Test of independence association between finger traits

Trait	Phenotype	Polydactyly		Hitchhiker's thumb		Bent little finger	
		Present	Absent	Single jointed	Double jointed	Straight	Bent
Mid-phalangeal hair	Present	1	2	40	16	54	2
	Absent	55	136	108	30	127	11
		$\chi^2 = 0.029$ p= 0.86 (NS)		$\chi^2 = 1.02$ p= 0.31 (NS)		$\chi^2 = 1.23$ p= 0.267 (NS)	
Polydactyly	Present	-	-	4	0	4	0
	Absent	-	-	144	46	177	13
Hitchiker's thumb	Single jointed	-	-	-	-	139	9
	Double jointed	-	-	-	-	42	4
						$\chi^2 = 0.383$ p= 0.536 (NS)	

Table 6. Test of independence association between hand and foot traits

Trait	Phenotype	Right handedness		Arched foot	
		Right	Left	Arched	Flat
Crossing of arms	Right over left	192	1	172	21
	Left over right	1	0	0	1
Right handedness		$\chi^2 = 0.0052$ p= 0.94		$\chi^2 = 7.85$ p= 0.005	
	Right	-	-	171	21
	Left	-	-	1	1
				$\chi^2 = 3.004$ p= 0.083 (NS)	

of crossing of arms (100% Christians, 99% Hindus and 100% Muslims) and right handedness (100% Christians, 99% Hindus and 98.1% Muslims). 73.6% Christians, 91.1% Hindus and 88.4% Muslims had an arched foot. The study found that there was no significant difference between the number of dominant and recessive characters expressed in all the three religions (Table-2).

3.3 Independent Association among the Estimated Traits in the Population of Study

3.3.1 Association analysis of tongue traits

A statistically significant relationship exists between the tongue rolling and tongue folding ($\chi^2 = 4.09$, $p = 0.05$, $df = 1$). The tendency for tongue roller to be a tongue folder was more prevalent in the sampled population (Table-3).

3.3.2 Association analysis of facial traits

The prevalence of widow's peak was independent of the other two facial traits, cleft chin ($\chi^2 = 0.566$, $p = 0.452$, $df = 1$) and earlobe attachment ($\chi^2 = 1.288$, $p = 0.256$, $df = 1$) (Table 4). But, only the frequency of widow's peak was significantly related with the frequency of dimpled cheeks ($\chi^2 = 4.80$, $p = 0.05$, $df = 1$). The prevalence of facial trait, cleft chin was independent of the other two traits, earlobe attachment ($\chi^2 = 1.13$, $p = 0.286$, $df = 1$) and dimpled cheeks ($\chi^2 = 1.54$, $p = 0.214$, $df = 1$). The prevalence of facial trait, earlobe attachment was independent of the trait, dimpled cheeks ($\chi^2 = 0.1903$, $p = 0.663$, $df = 1$).

3.3.3 Association analysis of finger traits

No relationship exists between selected finger traits (Table-5). Mid-phalangeal hair was independent with polydactyly ($\chi^2 = 0.029$, $p = 0.86$, $df = 1$), hitchhiker's thumb ($\chi^2 = 1.02$, $p = 0.31$, $df = 1$) and bent little finger ($\chi^2 = 1.23$, $p = 0.26$, $df = 1$). Polydactyly was independent with hitchhiker's thumb ($\chi^2 = 1.26$, $p = 0.259$, $df = 1$) and bent little finger ($\chi^2 = 0.293$, $p = 0.588$, $df = 1$). Hitchhiker's thumb has no relation with bent little finger ($\chi^2 = 0.383$, $p = 0.536$, $df = 1$).

3.3.4 Association analysis of hand and foot traits

Crossing of arms was independent with right handedness ($\chi^2 = 0.0052$, $p = 0.94$, $df = 1$), however, there was a strong correlation with arched foot ($\chi^2 = 7.85$, $p = 0.005$, $df = 1$) (Table-6). There was no relation with right handedness and arched foot ($\chi^2 = 3.004$, $p = 0.083$, $df = 1$).

4. DISCUSSION

4.1 Distribution of Dominant and Recessive Expressed Phenotypes of Morphogenetic Traits

The frequency distribution of tongue rolling and tongue folding capability in the sampled population specified that there were more tongue rollers and folders than non-rollers and non-folders. The present study was in validation with the studies of Eboh [20], Nwaopara et al. [21]; Odokuma et al. [22] and Adekoye et al. [23] who also observed the prevalence of more tongue rollers in their study. The frequency distribution of polydactyly in the sampled population revealed that majority of the individuals showed absence of polydactyly. Polydactyly, inspite of being a dominant morphogenetic trait is not expressed in the population as it is a congenital malformation and is not favourable to be expressed in the population. The present study conforms with the studies of Roberts et al. [24], Adekoye et al. [23] who reported the same results in their respective studies. The prevalence of Hitchhiker's thumb was found to be lower than straight or single jointed thumb in the present study which corresponds with the results of Onyije et al. [25] who observed lower prevalence of hitchhiker's thumb in his studied population. The present study slightly differs with the results of Usha et al. [4] who observed a higher prevalence of curved thumb (62.6%) in the females of Thrissur district in India. The study further showed that bent little finger had a lower prevalence than the straight phenotype among the population studied. The present study relates with the findings of Munir et al. [12] and Adekoye et al. [23]. The hitchhiker's thumb and bent little finger were expressed in a recessive manner and might be related to the evolutionary aspect of hand gripping. The frequency of widow's peak was less prevalent than the straight hairline which associates with the studies of Birla and Sinha [26], Ebeye et al. [17], Naz et al. [27] and Adekoye et al. [23]. The frequency of cleft chin in the present study was less compared to the round chin which is not in agreement with the studies of Omotoso et al. [28] and Razzaq et al. [29]. There was a very low frequency of dimpled cheeks (12.9%) in the present study than round cheeks which is in total agreement with the studies of Anibor et al. [30]. The presence or absence of the mid-phalangeal hair on the fingers can be attributed to the different environmental factors such as life-styles and nature of work. Majority of the individuals in the present study showed the absence of mid-phalangeal hair which was an autosomal recessive trait was more prevalent than presence of mid-phalangeal hair on the fingers which is in total agreement with the studies of Aboagye et al. [31], Pandey et al. [32] and Umoyen et al. [33]. Free

ear lobe is an autosomal dominant trait compared to the attached ear lobes which is an autosomal recessive trait. The frequency of occurrence of free ear lobes was more than the attached ear lobes in the present study which is in concurrence with the studies of Yadav et al. [34], Victor and Lopez [35] and Dongre [36]. The pattern of crossing of arms in the present study showed that 99.5% of the sampled population crossed their arms right over the left. It is observed that about 85% of the population all over the world [37]. About 99% of the sampled individuals in the present study showed right handedness. The results of the present study showed that the right handedness was one of the most dominant traits, illuminating a clear distinction between dominant and recessive trait in handedness and is in correlation with the studies of Chadha and Sandhu [38] and Usha et al. [4] from Indian populations. The lower representation of genetic disorders such as autism, allergies, alcoholism, neuropsychiatric bipolar disease and schizophrenia in the right handed individuals than the left handed individuals made right handedness as the dominant trait. In the present study, about 88.7% individuals showed the arched foot while the remaining 11.3% showed flat foot.

4.2 Distribution of Morphogenetic Traits amongst Religions

The study also revealed that there was no significant difference between the number of dominant and recessive characters expressed in the three religions studied. The expression of morphogenetic traits is influenced by various factors such as race, ethnicity, gender, nationality, environmental factors and genetic composition of the individuals [39]. Majority of the morphogenetic traits studies are under the control of single genes in a dominant or recessive manner with varying degree of expressivity and penetrance [4]. Population size or sample size is also one of the important factors as it affects the distribution frequencies and statistical differences within or among considered population. Also, the environmental factors such as life-style, occupation etc. affect the expression of morphogenetic traits. In the present study, the sampled population is female students and is involved in domestic chores and hence, the absence of mid-phalangeal hair is expressed as a dominant trait.

4.3 Independent Association among the Estimated Traits in the Population of Study

The majority of the studied morphogenetic traits such as bent little finger, cleft chin, dimple cheeks, hitchhiker's thumb, mid-phalangeal hair, polydactyly,

widow's peak were expressed in a recessive manner. These recessive phenotypes were more prevalent in the population of study. Remarkably, the dominant traits such as bent little finger, cleft chin, dimple cheeks, hitchhiker's thumb, mid-phalangeal hair, polydactyly, widow's peak are relatively less expressed in the population. Hence, it can be seen that dominant characters are not always expressed more in the population and expression of trait varies due to several factors.

4.4 Association Analysis of Tongue Traits, Facial Traits, Finger Traits and Hand and Foot Traits

The present study revealed that there is a significant association between tongue rolling and tongue folding as the tendency of tongue roller to be tongue folder was more prevalent. Among the facials traits, only dimpled cheeks and widow's peak showed a significant association while the other traits showed no association among the traits. Similarly, there was no significant association among the finger traits. Among the hand and foot traits, only there was a significant association between crossing of arms and arched foot while the traits have no association. These associations might be due to the proximity of genes controlling these morphogenetic traits. There was no association of morphogenetic traits with religion in the present study. There are only few association studies of morphogenetic traits with respect to gender (Odokuma et al. 2008; Munir et al. 2015; Adekoye et al. 2020) but the present study was carried out only in female students and the association of gender with the morphogenetic traits could not be analysed.

5. CONCLUSION

The results of this study has shown that no two individuals are similar and they differ in one more morphological and behavioural traits with each other. The expression of recessive traits such as bent little finger, cleft chin, dimpled cheeks, hitchhiker's thumb, mid-phalangeal hair, polydactyly, widow's peak was more prevalent among the sampled population than their dominant counterparts which implies that the morphogenetic traits express themselves based on several factors such as environmental, genetic drift, mutation etc. Similarly, some of the dominant traits of tongue rolling, tongue folding, earlobe attachment, right handedness, crossing of right arm over left and arched foot were observed more than their recessive counterparts in the sampled participants. Though the results of the present study were not typed to any ethnicity or group, associations of traits observed in the diverse population of students gave a crystal clear idea about the morphogenetic traits relation among

different religions in this area. These types of studies are of great importance as they can help in many subject areas such as forensic pathology, anthropology and population diversity studies in this region.

INFORMED CONSENT

Informed consent was obtained from all individual participants.

ETHICAL APPROVAL

All procedures performed in this study on human participants were in accordance with the ethical standards of the institutional research committee and with the Helsinki declaration and its later ethical approval.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

REFERENCES

1. Anna CP. Foundation of Genetics —a science for society, Tata McGraw-Hill Publishing Company Ltd, New Delhi. 1976:64-66.
2. Gandham B. Some Morphogenetic Traits among the Vennekula Kshatriya of Andhra Pradesh. *J Hum Ecol.* 1999;10: 313-314.
3. Chhikara SK, Yadav AS. Studies on morphogenetic and behavioural traits in five endogamous groups of Haryana. *The Asian Man - An Int Jour.* 2013;7:192-195.
4. Usha AU, Sidijo S, Stejin PG, Alisha KS, Anjana CP, Anju M. A study on the expression of some selected human morphogenetic traits in Thrissur district. *Sci.* 2016;12 (1): 94-100.
5. Singh J, Sengupta S. Some morphogenetic and behavioural traits among the Assamese Sikhs. *Anthropol.* 2014;6 (4):253-255.
6. Bhasin MK, Khanna A. Study of behavioural traits among nine population groups of Jammu and Kashmir. *J. Hum Ecol.* 1994;5:133-134.
7. Verma PS, Agarwal VK. Genetics, Eighth ed. S. Chand and Co. Ltd, New Delhi. *Sci.* 1989;100.
8. Das B, Sengupta S. A note on some morphogenetic variables among the Sonowal Kacharis of Assam. *Anthropol.* 2003;5(3):211-212.
9. Bama PS. Elucidating the linkage of blood groups with selected human morphogenetic traits in Dindigul of Tamil Nadu Ethnicity. *J. Adv. Scien. Res.* 2021;12 (1):103-107.
10. Liu TT, Hsu TC. Tongue folding and tongue rolling in a sample of the Chinese population. *J. of Heredit.* 1949;40:19-21.
11. Umair M, Ahmad F, Bilal M, Ahmad W, Alfadhel M. Clinical Genetics of Polydactyly: An updated review. *Front. Genet.* 2018;9 (447):1-9.
12. Munir S, Sadeeqa A, Nergis B, Tariq N, Sajjad N. Assessment of morphogenetic inherited traits: earlobe attachment, bent little finger and hitchhiker's thumb in Quetta, Pakistan. *World J. Zool.* 2015;10 (4):252-255.
13. Victor AM. Widow's Peak. in: Online Mendelian Inheritance in Man. Johns Hopkins University. 2009:194000.
14. Victor AM. Cleft Chin. In: Online Mendelian Inheritance in Man. Johns Hopkins University. 2013:119000.
15. Victor AM. Dimples, Facial. in: Online Mendelian Inheritance in Man, Johns Hopkins University. 1994;126100.
16. Westlund N, Oinonen KA, Mazmanian D, Bird JL. The value of middle phalangeal hair as an anthropometric marker: A review of the literature. *Homo.* 2015;66 (4):316-31.
17. Ebeye OA, Chris-Ozoko LE, Ogeneovo P, Onoriode A. A study of some morphogenetic traits among the Esan ethnic group of Nigeria. *East Afr. Med. J.* 2014;91 (11):420-422.
18. Wiener AS. Observations on the manner of clasping the hands and folding the arms. *Amer Natl.* 1932;66:365-370.
19. Gillian F, Caterina Q, David L, Denis M, Michael T. Human handedness: An inherited evolutionary trait. *Behavioural Brain Research.* 2012;237C:200-206.
20. Eboh DEO. Morphogenetic traits in eight South Nigerian ethnic groups. *Calabar J. Health Sci.* 2017;1(1):14-22.
21. Nwaopara AO, Anibeze CIP, Apkuaka FC, Agbontaen OF. Morphogenetic traits combination pattern among the population of Ekpoma, Nigeria: Focus on tongue rolling, earlobe attachment, blood groups and genotypes. *Afr. J. Biotechnol.* 2008;7 (20):3593-3598.
22. Odokuma IE, Eghworo O, Avwioro G, Agbedia U. Tongue rolling and tongue folding traits in an African population. *Int. J. Morphol.* 2008;26 (3):533-535.
23. Adekoye KO, Fakorede ST, Ogunkanmi AL, Amusa OD, Sifau MOOboh B. Inheritance pattern and association studies of some human morphogenetic traits among Nigerian

- undergraduate students. *Scientific African*. 2020;9:1-8.
24. Roberts FO, Waritimi GE, Dare NW. The prevalence of polydactyly in the Southern Ijaws of Bayelsa State, Nigeria. *Int. J. Basic Innov. Res.* 2018;7 (1):29-34.
25. Onyije FM, Oyinbo CA, Waritimi EG. The prevalence and comparison of bent little finger and hitchhiker's thumb in South-South Nigeria. *Eur. J. Appl. Sci.* 2012;4 (4):157-159.
26. Birla S, Sinha KK. Analysis of morphogenetic traits among various endogamous human population groups. *Ind J. Appl. Res*, 2019; 9(3):71-73.
27. Naz G, Hameed F, Bibi Z, Khatoon D, Sajjad N. Hairline shapes: study of a morphogenetic inheritance trait. *Pak. J. Biochem. Mol. Biol.* 2014;47 (1-2):118-120.
28. Omotoso GO, Adeniyi PA, Medubi LJ. Prevalence of facial dimples amongst South-Western Nigerians: A case study of Ilorin, Kwara State of Nigeria. *Int. J. Biomed. Health Sci.* 2010;6(4):241-244.
29. Razzaq R, Kanwal SS, Tariq N, Sajjad N. Tongue rolling, folding, cheek dimple and chin cleft: study of a morphogenetic traits in Quetta population. *World J. Zool.* 2015;10(3): 237-240.
30. Anibor E, Ogbor-Omorie E, Opiah AE. The distribution of widow's peak trait among young adults of the Isoko ethnic group in Delta State, Nigeria, *Ann. Bioanthropol.* 2015;3:47-49.
31. Aboagye KM, Tsegah AM, Ussif. Distribution of hairs on the phalanges of hands among Ghanaians. *J. Anthropol*; 2013.
32. Pandey BN, Jahangeer MD, Priyanka M. A morphogenetic study of Badhiya Muslims of Purnia District (Bihar), India. *Int. J. Life Sci.* 2013;1 (3):233-238.
33. Umoyen AJ, Akpan NG, Abu GI, Thomas TL, Uyokei U. The inheritance pattern of some human morphogenetic and serological traits among two Nigerian ethnic groups in Akwa-Ibom State. *Sch. Int. J. Anat. Physiol.* 2021;4(6):65-74.
34. Yadav JS, Yadav AS, Chadha PC. Studies on morphogenetic and behavioural traits in five endogamous groups of Harayana. *J. Pan. Afr. Stud.* 2000;2:329-332.
35. Victor AM, Lopez A. Earlobe Attachment, Attached vs. Unattached. in: *Online Mendelian Inheritance in Man*, Johns Hopkins University. 2010:128900.
36. Dongre S. Morphogenetic expressions, study in Muslim population. *Thin. Ind. Journ.* 2019;22(31):518-523.
37. Marchant LF, McGrew WC. Human handedness: An ethological perspective. *Hum. Evol.* 1998;13: 221-228.
38. Chadha P, Sandhu SK. A study on distribution of morphological, behavioural and serological traits in three endogamous groups of scheduled castes in Jammu and Kashmir. *Cibtech J. Zool.* 2013;2(1):47-50.
39. Naik SN, Ranjith Y. A survey of human traits with simple traits with simple inheritance patterns in Sahayadri Science College Campus Shivamogga. *Int J. Scien Res Rev.* 2019;8(2): 102-109.

(APPENDIX-I)

QUESTIONNAIRE SHEET

Please provide your response to each items included in the Questionnaire sheet. Your response will be kept confidential and used for the academic research purpose only.

I give consent to participate in this survey and has no objection to provide the data for Research purpose.

Signature

Name of the participant:

College/University Course):

Age:

Religion:

Genetic traits

Put Tick mark against appropriate trait

Rolling of tongue	Roller	Non-rollers
Folding of tongue	Folder	Non-folders
Polydactyly	Present	Absent
Hitchhiker's thumb (Curved thumb)	Single jointed	Double jointed
Bent little finger	Straight	Bent
Widow's Peak	Curved hairline	Straight hairline
Cleft Chin	Cleft	Round
Dimpled cheeks	Present	Absent
Mid-phalangeal hair	Present	Absent
Earlobe attachment	Attached	Free
Crossing of arms	Present	Absent
Right handedness	Right	left
Arched foot	Arched	Flat